

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Dental Record, Treatment and Appointment System

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Group Members:
Robul Grantosa
Reinghnier Dale Doronila
Van Vincent Lopez
France Mary Nicdao
Krystel Kate Somera

System Development Track

Second Semester
2025-2026

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In recent years, the healthcare industry has increasingly adopted digital technologies to improve service delivery, data management, and patient engagement. Dental clinics, however, still rely heavily on manual processes such as paper-based patient records and traditional appointment scheduling systems. These methods often lead to inefficiencies, data redundancy, misplacement of records, and scheduling conflicts.

In addition, manual record-keeping makes it difficult for dental professionals to quickly retrieve patient histories, monitor treatment progress, and ensure accurate documentation. This can result in delays in service, reduced productivity, and potential errors in patient care. Patients may also experience inconvenience due to long waiting times, lack of appointment reminders, and limited access to their own dental information.

With the rapid growth of mobile and web technologies, there is a strong need to modernize dental clinic operations through integrated digital solutions. A system that combines both mobile accessibility for patients and a web-based platform for dentists and administrators can significantly enhance workflow efficiency and service quality. Such systems enable real-time data access, automated scheduling, and improved communication between patients and healthcare providers.

Moreover, the implementation of a digital system promotes better data security, organization, and backup compared to traditional paper-based methods. It also supports data analytics, allowing clinics to track patient trends, treatment outcomes, and overall clinic performance. This can help in making informed decisions and improving long-term service strategies.

The proposed **Dental Record, Treatment and Appointment System** aim to address these challenges by providing a centralized platform for managing patient information, tracking treatment history, and scheduling appointments. By incorporating features such as digital patient records, online appointment booking, and dental chart visualization, the system seeks to improve accuracy, accessibility, and overall patient care.

Furthermore, the system is designed to enhance user experience for both clinic staff and patients by offering a user-friendly interface, efficient data processing, and reliable system performance. Ultimately, this study aims to contribute to the advancement of dental clinic management through the adoption of modern technology, leading to improved operational efficiency and higher patient satisfaction. Additionally, the system supports accurate record management, faster appointment scheduling, secure data storage, and improved communication between patients and dental professionals. This helps clinics provide more organized, accessible, and high-quality healthcare services while reducing administrative workload and operational delays.

1.2 Statement of the Problem

General Problem

Dental clinics experience inefficiencies in managing patient records, treatment tracking, and appointment scheduling due to reliance on manual and fragmented systems. This often leads to delays in retrieving information, increased risk of data errors, miscommunication among staff, and difficulty in maintaining accurate and updated records. It also affects overall workflow efficiency and reduces the quality of patient service and clinic management. In addition, these challenges can result in missed appointments, duplication of work, and limited access to real-time patient information, which further impacts the effectiveness of dental health care delivery.

Specific Problems

1. How can patient information be efficiently stored and managed in a digital format?
2. How can treatment history and dental records be accurately tracked and easily accessed?
3. How can patients conveniently schedule, reschedule, or cancel appointments online?

4. How can dentists effectively visualize and assess a patient's dental condition?
 5. How can the system reduce manual errors and improve overall clinic workflow efficiency?
-

1.3 Objectives of the Study

General Objective

To develop a **Dental Record, Treatment and Appointment System** that enhances the efficiency, accuracy, and accessibility of dental clinic operations through a mobile and web-based platform.

Specific Objectives

1. To design a system that enables digital registration and management of patient information.
 2. To implement a feature for tracking and storing treatment history and dental records.
 3. To develop an online appointment scheduling system with booking, rescheduling, and cancellation functionalities.
 4. To create an interactive dental chart visualization for accurate diagnosis and treatment planning.
 5. To improve clinic workflow by reducing manual processes and minimizing errors.
-

1.4 Scope and Delimitation

Scope

The study focuses on the design and development of a **Dental Record, Treatment and Appointment System** with two main platforms:

1. **Mobile Application (User Side):**

- Patient registration and profile management
- Viewing and managing dental records
- Appointment scheduling and management

2. **Web Application (Admin/Dentist Side):**

- Patient record management
- Appointment monitoring
- Dental chart visual mapping with interactive features

Delimitation

1. The system is limited to dental clinic operations only.
 2. Payment processing and billing features are not included.
 3. The system requires internet connectivity for full functionality.
 4. Advanced AI-based diagnosis is not included in the system.
 5. The system is designed for a single clinic setup and not for multi-branch integration.
-

1.5 Significance of the study

This study provides benefits to the following:

1. **Dental Clinics:**

Improves efficiency in managing patient records, appointments, and treatment monitoring.

2. **Dentists:**

Provides tools for better diagnosis through visual dental charts and organized patient data.

3. **Patients:**
Offers convenience through mobile access to records and easy appointment scheduling.
 4. **Clinic Administrators:**
Reduces paperwork and enhances data organization and retrieval.
 5. **Future Researchers:**
Serves as a reference for developing similar healthcare management systems.
-

1.6 Definition of Terms

Dental Record: A digital file containing patient information, treatment history, and dental images.

Appointment Scheduling: The process of booking, managing, and organizing patient visits.

Dental Chart: A visual representation of a patient's teeth used for diagnosis and treatment planning.

Mobile Application: A software application designed to run on smartphones for patient use.

Web Application: A browser-based system used by dentists and administrators to manage clinic operations.

Patient Management: The process of handling patient data, records, and history.

Treatment History: A record of past dental procedures performed on a patient.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Recent foreign studies highlight the growing importance of digital technologies in healthcare systems. Health Information Technologies (HIT), including electronic health systems, mobile health applications, and cloud-based platforms, have transformed how healthcare services are delivered, managed, and evaluated. These technologies improve healthcare efficiency, patient engagement, and clinical decision-making.

A study by Barker et al. (2024, pp. 2–5) found that modern health information technologies, including mobile and web-based systems, significantly improve patient-centered care by enhancing accessibility, data management, and communication between healthcare providers. These technologies allow healthcare providers to deliver more efficient and coordinated services. Also, these systems reduce manual errors and improve efficiency in clinical workflows, allowing healthcare professionals to focus more on patient care rather than administrative tasks.

Similarly, Shen et al. (2025, pp. 4–8) emphasized that Electronic Health Record (EHR) systems improve interoperability and facilitate better data sharing among healthcare providers, resulting in improved treatment outcomes and clinical decision-making. This integration results in better diagnosis accuracy, faster treatment planning, and improved patient safety.

Furthermore, Bogale et al. (2024, pp. 3–6) conducted a systematic review and found that electronic records significantly improve healthcare service delivery and reduce mortality risks by enhancing data accessibility, clinical accuracy, and communication among healthcare professionals. Their findings revealed that healthcare facilities using electronic systems experienced better patient monitoring and more organized clinical operations.

In addition, Kruse et al. (2018, pp. 1–8) explained that electronic medical systems improve operational efficiency by minimizing paperwork, reducing redundant medical procedures, and increasing the accuracy of patient information management. The researchers also stated that digital healthcare systems contribute to improved patient satisfaction because services become faster, more organized, and more accessible.

Another study by Campanella et al. (2019, pp. 60–64) revealed that implementing electronic health systems significantly improves healthcare quality by reducing medical errors, improving documentation, and supporting evidence-based medical practices. The study further highlighted that digital systems increase the efficiency of healthcare professionals through streamlined workflows and automated processes.

These studies demonstrate that integrating digital systems into healthcare environments leads to improved efficiency, better patient outcomes, enhanced service quality, and more effective management of patient information.

2.2 Local Studies

In the Philippine context, digital health adoption continues to grow as healthcare institutions transition toward more efficient and technology-driven systems. Recent studies emphasize the importance of adopting digital health systems and implementing electronic systems in both urban and rural healthcare facilities.

Acacio-Claro et al. (2024, pp. 7–10) examined the adoption of Electronic Medical Records (EMRs) in rural health facilities in the Philippines and found that digital systems improve record management, data accessibility, reduce paperwork, and enhance service delivery efficiency. However, they also noted challenges such as limited infrastructure, lack of training, and resistance to technology adoption.

Derecho et al. (2024, pp. 5–9) analyzed EMR adoption in developing economies and highlighted that digital systems enhance healthcare efficiency, although challenges such as system usability and infrastructure still exist. Despite these challenges, the study concluded that digital transformation remains essential for improving healthcare accessibility in the Philippines.

Additionally, Marcelo et al. (2018, pp. 15–20) emphasized that the implementation of health information systems in the Philippines supports better healthcare coordination and data management. The study highlighted that digital systems help healthcare providers improve service delivery by ensuring faster retrieval of patient information and reducing errors caused by manual recordkeeping.

Moreover, Garcia et al. (2021, pp. 25–30) found that appointment scheduling systems in local clinics significantly reduce patient waiting times and improve clinic workflow efficiency. The researchers explained that automated appointment systems help healthcare facilities manage patient schedules more effectively, resulting in better time management and improved patient satisfaction.

These local studies confirm that digital healthcare systems are increasingly relevant in the Philippines and play a vital role in improving healthcare services.

2.3 Related Literature

2.3.1 Electronic Health Records (EHRs)

Electronic Health Records (EHRs) are essential tools for modern healthcare systems. Atasoy et al. (2019, pp. 487–490) explained that EHR adoption improves healthcare quality by reducing medical errors, reduces costs, and enhances clinical decision-making by providing accurate and accessible patient data.

Recent studies also indicate that EHR systems support interoperability among hospitals, laboratories, and clinics, and data sharing across healthcare providers to access complete patient histories in real time. This leads to more accurate diagnosis, more coordinated, efficient care, and improved treatment outcomes (Shen et al., 2025, pp. 6–9).

EHR adoption has also been associated with increased efficiency in healthcare workflows by reducing duplication of laboratory tests, minimizing paperwork, and improving data organization. Healthcare providers can retrieve patient information more quickly, allowing them to provide timely and effective healthcare services. Furthermore, electronic records improve data security and backup management. Unlike paper-based systems that are

vulnerable to physical damage or loss, electronic systems provide safer storage through cloud databases and encrypted platforms. This ensures that patient records remain secure, confidential, and easily recoverable when needed.

2.3.2 Mobile Health Applications (mHealth)

Mobile health (mHealth) applications have become increasingly important in modern healthcare delivery, improving patient engagement and healthcare access. Arigo et al. (2024, pp. 122–126) found that mobile health technologies enhance patient participation, improve communication with healthcare providers, and support better health outcomes.

These applications enable patients to access their consultation, health information, medical records, receive notifications, and manage appointments conveniently, making them essential in modern healthcare systems.

Additionally, mHealth systems support preventive healthcare by encouraging healthy lifestyles and continuous health monitoring. Through mobile notifications and health tracking features, patients become more aware of their health conditions and treatment progress. This contributes to improved patient responsibility and long-term healthcare management. Mobile applications also provide convenience for healthcare providers by enabling faster communication and improving patient monitoring processes. In dental clinics, mobile-based systems can help patients receive treatment updates, reminders for dental checkups, and notifications regarding follow-up procedures.

2.3.3 Appointment Scheduling Systems

Digital appointment scheduling systems improve healthcare efficiency by reducing patient waiting time and optimizing clinic workflows. Recent studies show that automated scheduling systems enhance clinic workflow by optimizing time management and reducing missed appointments (Barker et al., 2024, pp. 6–9).

Online scheduling platforms allow patients to schedule, reschedule, or cancel appointments without physically visiting the clinic. This improves convenience, accessibility, and overall patient satisfaction. Automated systems also reduce the burden on clinic staff because appointment records are managed electronically and updated in real time.

Moreover, appointment scheduling systems help clinics avoid overcrowding and improve daily operational planning. Healthcare providers can monitor patient schedules efficiently, allocate consultation time properly, and reduce scheduling conflicts.

Automated reminders sent through SMS or mobile notifications further improve appointment attendance and reduce no-show rates. These systems enhance communication between patients and healthcare providers while ensuring more organized clinic operations.

2.3.4 Dental Visualization and Charting Systems

Digital visualization tools and charting systems play a crucial role in modern dental care. Recent literature indicates that digital charting systems improve diagnostic accuracy and enable better treatment planning by providing visual representations of dental conditions (Shen et al., 2025, pp. 10–12).

These systems allow dentists to document oral conditions, tooth status, treatment procedures, and patient progress more accurately. Digital dental charts also help monitor changes in oral health over time, improving long-term patient care management.

Additionally, visualization tools improve patient understanding by allowing dentists to explain dental conditions and treatment procedures more clearly through images and digital diagrams. Patients become more informed about their oral health conditions, increasing their confidence and cooperation during treatment.

Digital dental charting systems also improve record organization and reduce the risk of losing patient information. Since records are stored electronically, dental clinics can easily retrieve treatment histories, monitor completed procedures, and generate reports for future consultations.

2.4 Synthesis of the Reviewed Literature

The reviewed literature from 2018 to the present highlights the significant role of digital technologies in improving healthcare services. Electronic Health Records (EHRs) enhance data accuracy, accessibility, and interoperability, and clinical decision-

making, leading to better clinical outcomes. Mobile health applications improve patient engagement and accessibility.

Appointment scheduling systems contribute to improved clinic efficiency by reducing waiting times and optimizing workflow management. Additionally, digital visualization tools support accurate diagnosis and effective treatment planning in dental care settings.

Foreign studies emphasize the global benefits of integrating digital healthcare systems integration, including improved efficiency, safety, and patient outcomes. Local studies, particularly in the Philippines, confirm their relevance and effectiveness of these technologies but also highlight challenges such as infrastructure limitations, training gaps, and system adoption barriers. These findings justify the development of the proposed Dental Record, Treatment and Appointment System, as it integrates key features such as digital records, mobile accessibility, appointment scheduling, and visualization tools to address current challenges in dental clinic management. These proposed system aims to improve efficiency, accessibility, and quality of dental healthcare services.

The reviewed literature supports the development of the proposed **Dental Record, Treatment and Appointment System** because it integrates essential healthcare technologies such as electronic records management, appointment scheduling, treatment monitoring, and digital accessibility. The proposed system aims to address common challenges in dental clinic management, including manual recordkeeping, inefficient scheduling, data retrieval difficulties, and poor patient monitoring

Therefore, the proposed system is expected to improve operational efficiency, enhance patient service quality, strengthen data management, and provide a more organized and accessible dental healthcare experience for both patients and dental professionals.

References:

Acacio-Claro, P. J., et al. (2024). Adoption of electronic medical records in rural health facilities in the Philippines. *BMC Medical Informatics and Decision Making*, 24(1), 1–12.

Arigo, D., Jake-Schoffman, D. E., & Hekler, E. B. (2024). The future of mobile health applications and patient engagement. *Journal of Behavioral Medicine*, 47(2), 120–130.

Atasoy, H., Greenwood, B. N., & McCullough, J. S. (2019). The digitization of patient care: A review of the effects of electronic health records on health care quality and costs. *Annual Review of Public Health*, 40, 487–500.

Barker, W., Johnson, C., & Smith, L. (2024). Advancing patient-centered care through digital health technologies. *Journal of Medical Internet Research*, 26, e59791.

Bogale, S., Tadesse, T., & Endalamaw, A. (2024). The impact of electronic health records on healthcare quality: A systematic review. *BMC Health Services Research*, 24(1), 1–10.

Campanella, P., Lovato, E., Marone, C., et al. (2019). The impact of electronic health records on healthcare quality: A systematic review and meta-analysis. *European Journal of Public Health*, 26(1), 60–64.

Derecho, A. M., et al. (2024). Electronic medical record adoption in developing countries: Challenges and opportunities. *International Journal of Medical Informatics*, 183, 105308.

Garcia, R. T., Santos, M. L., & Reyes, P. D. (2021). Implementation of appointment scheduling systems in local healthcare clinics. *Philippine Journal of Health Research and Development*, 25(2), 25–30.

Kruse, C. S., Kristof, C., Jones, B., Mitchell, E., & Martinez, A. (2018). Barriers to electronic health record adoption: A systematic literature review. *Journal of Medical Systems*, 42(3), 1–8.

Marcelo, A. B., et al. (2018). Health information systems in the Philippines: Challenges and opportunities. *Acta Medica Philippina*, 52(1), 15–20.

Shen, Y., Zhang, X., & Liu, H. (2025). Enhancing healthcare interoperability through electronic health records: A global perspective. *Journal of Medical Internet Research*, 27, e59024.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study utilizes a developmental research design, which focuses on the design, development, and evaluation of a system intended to solve identified problems. The approach is appropriate because the study aims to create a Dental Record, Treatment and Appointment System that improves patient record management, appointment scheduling, and dental diagnosis processes. Traditional methods, such as manual paper-based record-keeping, fixed appointment systems, and disconnected documentation of treatment procedures, often result in issues like lost or misplaced records, scheduling conflicts, slow access to patient medical history, inconsistent documentation, and inefficient workflow management. This design provides the clear, step-by-step structure needed to develop a solution that directly overcomes these difficulties and improves overall clinic operations.

The developmental design allows the researchers to systematically plan, build, test, and evaluate the system to ensure its functionality, usability, and effectiveness in a real-world dental clinic setting. It follows a logical progression that begins with identifying the specific needs of users including patients, dentists, and administrative staff and moves through the stages of conceptualization, design, implementation, and assessment. By using this approach, every component of the system is developed with clear purpose and relevance to daily clinical activities. This method ensures that the final product not only meets technical specifications but also operates reliably and efficiently in actual working environments. It allows the team to verify that the system successfully achieves its core goals: improving the accuracy, security, and accessibility of patient records; streamlining the entire process of booking, rescheduling, and managing appointments; simplifying the documentation and tracking of dental treatments and procedures; and ultimately enhancing the quality of patient care and the overall productivity of the dental facility.

Furthermore, this research design is highly suitable because it emphasizes continuous improvement and validation throughout the development cycle. It enables the researchers to gather feedback from potential users at various stages, allowing for adjustments and refinements that align the system more closely with actual operational requirements. Unlike other research designs that may only focus on observation or theoretical analysis, the development approach is action-oriented and solution-driven, making it ideal for creating a functional information system. It ensures that the study does not merely identify problems but actively constructs, tests, and perfects a digital solution.

Adopting a developmental research design supports the creation of a system that is scalable and adaptable to future needs. During the planning and development phases, researchers can anticipate potential growth in patient volume, expansion of clinic services, or changes in regulatory requirements, and incorporate flexibility into the system's architecture. This approach ensures that the development platform remains relevant and useful over time, rather than becoming obsolete quickly. This strengthens the significance and contribution of the study to the broader field of health informatics and dental service management.

3.2 System Development Methodology

The study adopts the Agile Software Development Methodology, which emphasizes iterative development, continuous feedback, and flexibility.

Phases of Agile Methodology:

1. Planning Phase

In this phase, system requirements were identified based on the needs of patients, dentists, and clinic administrators. Key features such as patient management, appointment scheduling, and dental chart visualization were defined.

2. Design Phase

The system architecture, user interface, and database structure were designed. Wireframes and prototypes were created for both mobile and web applications to ensure user-friendly navigation.

3. Development Phase

The system was developed in iterations. Features were implemented gradually, starting with core functionalities such as registration and appointment scheduling, followed by advanced features like dental chart visualization.

4. Testing Phase

Each module was tested to identify and fix errors. Functional and usability testing were conducted to ensure system reliability.

5. Deployment and Evaluation Phase

The system was deployed in a test environment and evaluated based on performance, usability, and functionality.

3.3 System Architecture

The system follows a client-server architecture, consisting of two main components:

1. Mobile Application (Client Side)

Used by patients

Handles user input such as registration, appointment booking, and record viewing

Communicates with the server through API requests

2. Web Application (Admin/Dentist Side)

Used by dentists and administrators

Manages patient data, appointments, and dental charts

Provides tools for diagnosis and treatment tracking

3. Server (Backend System)

Processes requests from both mobile and web applications

Handles business logic, authentication, and data storage

Connects to the database for storing and retrieving information

4. Database

Stores patient records, appointment details, treatment history, and uploaded images

Ensures centralized and secure data management

3.4 Tools and Technologies Used

The development of the system utilized the following tools and technologies:

Programming Languages and Frameworks:

1. React.js – for developing the web-based admin/dentist interface
2. React Native – for building the mobile application for patients
3. Express.js – for creating the backend server and API services

Development Environment:

1. Visual Studio Code (VS Code) – used as the primary code editor

Database:

1. Firebase – a cloud-based platform used for storing and managing patient records, appointment details, treatment history, and uploaded images in real time

Other Tools:

1. Git for version control

These tools and technologies were selected for their efficiency, scalability, and compatibility in developing modern web and mobile applications.

3.5 Data Gathering Techniques

To ensure that the proposed system meets the specific needs of the dental clinic, the following methodologies were employed:

- **Document Analysis:** A comprehensive review of existing dental clinic records was conducted to understand how information is currently recorded, stored, and managed. These documents included patient information forms, dental charts, treatment records, consent forms, and appointment logbooks. This analysis helped identify the necessary data fields, workflow processes, and documentation structure required for the development of the proposed digital system. Document analysis helped reveal inconsistencies, redundancies, and limitations in the manual record-keeping system. It provided a basis for designing a structured and standardized database schema for the proposed digital system. This method also ensured that all necessary legal and clinical documentation requirements were considered in the system design.
 - **Semi-Structured Interviews:** Qualitative interviews were carried out with dentists, dental assistants, and clinic receptionists. This method allowed the researchers to prepare guided questions while also giving participants the flexibility to share their experiences, opinions, and suggestions in detail. The interviews focused on identifying common challenges such as difficulty in retrieving patient records, risks of data loss, scheduling conflicts, and time-consuming administrative tasks. This technique provided valuable qualitative data that helped define the functional and non-functional requirements of the system. It also ensured that the perspectives of different user roles were considered in the system development process.
 - **Observation:** Direct observation of daily clinic operations was performed to examine how patient records are handled, treatments are recorded, and appointments are scheduled and managed. This method provided insights into workflow inefficiencies, time delays, and automation could improve overall clinic performance. Observation provided contextual understanding that may not be fully captured through interviews alone, such as staff behavior, time management, and real-time problem-solving practices. These insights were essential in designing a system that not only meets functional requirements but also enhances overall workflow efficiency and user experience.
-

3.6 Testing Procedures

- **Unit Testing:** It was performed on individual components of the system to verify that each part functions correctly in isolation. Key modules tested included patient record management, treatment recording, appointment scheduling, and user authentication. Each unit was tested using predefined test cases to ensure that inputs, processes, and outputs

behaved as expected. For example, the patient record module was tested for accurate data entry, editing, and retrieval, while the appointment module was evaluated for correct date and time handling. This phase helped detect and fix errors early in the development process, reducing the risk of larger system failures later on. Unit testing plays a fundamental role in building a strong foundation for the entire system, as it allows developers to validate the functionality of every basic function before they are combined with other components. By testing each module separately, the team can isolate specific areas of the system and confirm that they perform exactly as required, which is especially important for sensitive functions like user authentication and data management. For instance, beyond basic data handling, the patient record module was also tested for features such as secure data storage, search functionality, and compatibility with different data formats, while the appointment module was checked for rules like conflict detection, reminder settings, and status updates. This level of detailed testing ensures that even the smallest operations work flawlessly, preventing potential errors that could disrupt daily clinic tasks and ensuring that the system operates smoothly at its most basic level.

- **System Testing:** The complete system was tested in a controlled environment to evaluate its overall performance, functionality, and reliability. This includes checking for errors, system response time, and proper handling of user inputs. System testing assessed the system's ability to handle multiple users and maintain data integrity. Any identified issues, such as slow response times or unexpected errors, were documented and addressed before proceeding to the next phase. System testing moves beyond individual components to examine how all modules work together as a unified platform, ensuring that interactions between different parts of the system are seamless and efficient. This phase simulates the actual workload and usage patterns that the system will face in a real dental clinic, allowing testers to observe how the platform performs under various conditions. In addition to checking for general errors and response speeds, the testing process also evaluated how the system manages important aspects such as data security, backup and recovery processes, and consistency of information across all features. It also tested scenarios where different users access different functions at the same time, verifying that the system remains stable, data remains accurate and secure, and that all operations continue to run smoothly even when the system is being used actively by multiple people. This ensures that the platform can support the high demands of daily clinic operations without compromising quality or reliability.
- **User Acceptance Testing:** It was carried out to determine whether the system meets the expectations and requirements of its intended users. The system was presented to intended users, including dentists, clinic staff, and receptionist, to evaluate its usability and effectiveness. This ensures that the final system is designed around the needs of its users, making it easier to adopt, more effective in practice, and ultimately more successful in improving the management of dental records, treatments, and appointments.

3.7 Evaluation Criteria

Metric	Target Benchmark
Response Time	High-priority actions such as saving patient records, updating treatment plans, or booking appointments must be processed and displayed within 3 seconds of submission.
Concurrent User Support	The system must remain stable, accurate, and fully functional while supporting at least 50 simultaneous users (dentists, staff, and patients) performing different tasks at the same time.
System Uptime	The platform must be accessible and operational throughout regular clinic hours, with minimal downtime; any maintenance must be scheduled outside service hours to avoid disruption.
Data Accuracy and Integrity	All stored data including personal details, treatment records, and appointment schedules must be correct, consistent, and complete with no duplication or loss of information.
Data Security and Privacy	Only authorized personnel can access, edit, or share patient records; all sensitive information must be protected and stored securely to prevent unauthorized access or leaks.
Record Retrieval Speed	Searching, viewing, or pulling up patient dental records, treatment history, or appointment logs must be done within 3 seconds, even with large volumes of stored data.
Usability	The interface must be easy to learn and navigate; users should be able to complete core tasks like adding records or scheduling appointments in 3 steps or less, with clear instructions and labels.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD - Level 0)

User → Dental System → Database

- Dentist/Staff inputs patient information, treatment details, and appointment schedules.
 - System processes, stores, and updates records in the database.
 - Authorized users retrieve, search, or generate reports from stored data.
-

Entity Relationship Diagram (ERD)

Entities:

- Patient (PatientID, Name, Age, Contact Info, Medical History, Registration Date)
- Dental Record (RecordID, PatientID, Diagnosis, Treatment Done, Notes, Date Created)
- Appointment (AppointmentID, PatientID, DentistID, Date, Time, Purpose, Status)
- Dentist/Staff (UserID, Name, Role, Contact, Login Credentials)

Relationships:

- Patient has one or more Dental Records
 - Patient schedules one or more Appointments
 - Dentist/Staff manages records and handles appointments
-

Use Case Diagram

The Use Case Diagram outlines the different users or actors interacting with the system and the specific functions or actions they can perform, organized according to their roles and responsibilities. It serves as a comprehensive visual and descriptive representation of how the **Dental Record, Treatment and Appointment System** operates, detailing every interaction between users and the platform to ensure all

operational, clinical, and administrative needs of a dental clinic are fully addressed. By clearly defining what each user can do, this diagram establishes the system's complete scope, aligning all features with the main goal of improving record management, treatment tracking, and appointment scheduling while enhancing overall service quality and patient satisfaction.

Additionally, it helps developers and stakeholders understand system boundaries, identify user requirements, and ensure proper workflow design. It also supports system validation and reduces design errors by clearly mapping interactions between patients, dentists, and administrative staff within the system. Furthermore, it provides a foundation for system development, testing, and future enhancements by ensuring that all functional requirements are properly documented and understood before implementation.

Actors:

● Patient

Patients are the primary beneficiaries and end-users of the system, utilizing it to manage their personal dental care journey conveniently and efficiently. Through the platform, they can create and maintain their own accounts, update personal details whenever necessary, and access their complete dental profile including past diagnosis, treatment history, and follow-up instructions. They play a key role in the scheduling process by booking, rescheduling, or canceling appointments based on their own availability and the clinic's open slots, while also receiving automated reminders to avoid missed visits. This role empowers patients to actively participate in their oral health management, reduces uncertainty about their treatment plans, and eliminates the inconvenience of manual inquiries or long waiting times, ensuring a smoother and more transparent experience throughout their engagement with the dental clinic. Patients can receive notifications regarding appointment changes, view clinic updates, and access important dental health information through the system, ensuring continuous engagement and better awareness of their ongoing treatment and oral health status. They also communicate with dental professionals when needed, helping them clarify concerns and better understand and prescribed procedures, which further improves trust, cooperation, and overall treatment outcomes.

● Dentist/Clinic Staff

Dentists and authorized clinic staff serve as the core operational users responsible for managing clinical information, delivering services, and maintaining daily workflow within the system. They are granted full access to create, view, update, and secure all patient-related data, including personal profiles, detailed dental records, prescribed treatments, medications administered, and clinical notes. They oversee the entire appointment system, managing schedules, confirming bookings adjusting time slots, and resolving scheduling conflicts to ensure the clinic operates smoothly and efficiently. Additionally, they use the system to search and retrieve records

instantly, track treatment progress, prepare necessary documentation, and generate reports that support treatment planning and service delivery. Their role is essential in ensuring that all data is accurate, complete and properly organized, directly contributing to better clinical decision-making, consistent quality of care, and effective communication between the clinic and its patients. Furthermore, they can monitor patient histories for better diagnosis accuracy, coordinate with other staff members in real time, and ensure proper follow-up care is provided, improving overall clinic efficiency and patient treatment outcomes. They can also review logs for system accountability, manage system queues effectively, and utilize system-generated insights to improve workflow of planning and enhance overall clinic performance.

- **Administrator**

The Administrator acts as the central authority and system manager, tasked with overseeing the entire platform, maintaining its security, and ensuring all operations run reliably and in compliance with privacy and data protection standards. This role involves creating and managing user accounts, assigning specific roles and access permissions based on job function, updating system settings, and monitoring user activity to prevent unauthorized access or misuse of sensitive information. The Administrator is also responsible for data backup, system maintenance, and ensuring that all records are stored securely and retrievable at all times. By controlling access levels and enforcing security protocols, this role safeguards confidential patient details, protects the integrity of the database, and ensures that the system remains stable, functional, and fully aligned with the clinic's operational policies and legal requirements. This oversight is critical to building trust, maintaining data privacy, and supporting the long-term effectiveness of the system.

Use Cases:

- **Register Patient Account**

This use case allows new patients or clinic staff to create a formal record in the system by entering essential personal details such as full name, contact number, address, and medical history. It ensures that every individual is properly registered with a unique identification number, which serves as the basis for all future transactions, record-keeping, and scheduling activities within the system. This registration process is fundamental because it establishes the basic foundation for managing every aspect of the patient's relationship with the clinic from scheduling visits and tracking treatments to storing clinical records and generating reports. By formalizing this process, the system ensures that every individual is properly identified, that their information is organized systematically, and that there is no duplication or confusion in records, making all subsequent operations faster, more accurate, and more efficient.

- Add/Update Dental Record

This function is exclusively available to dentists and authorized staff, enabling them to create new dental records or modify existing ones. It includes entering diagnosis results, treatment procedures performed, prescribed medications, clinical notes, and follow-up recommendations. This ensures that all dental information is accurately documented, kept up-to-date, and securely stored for easy reference during future visits or consultations.

Having reliable and current records available allows medical professionals to refer back to past information easily during future visits or consultations, ensuring that treatment is consistent, informed, and based on the most recent and relevant data available. In addition, this function helps reduce documentation errors and improves the overall accuracy of patient information by allowing real-time updates during or immediately after consultations. It also supports better coordination among dental staff by ensuring that all authorized users are working with the same patient records. It enhances decision-making by providing a complete and continuous history of a patient's dental condition and treatments over time.

- Schedule/Reschedule/Cancel Appointment

This use case manages the entire process of booking and managing appointments, providing a complete set of tools that allow both patients and clinic staff to set up new visits, change existing arrangements, or cancel appointments entirely. The system displays the current availability of time slots, specific dentists or service providers, and other relevant details, so users can make choices based on what is open and suitable for them. It helps prevent common problems such as double-booking or overlapping schedules by automatically updating the calendar and reflecting changes in real time. If adjustments are needed whether due to a patient's request, a change in the dentist's availability, or unforeseen circumstances the system allows for easy rescheduling or cancellation, and it can send instant notifications to all relevant parties to inform them of the changes. This function ensures that the clinic's daily operations run smoothly and efficiently, reduces conflicts and confusion, and gives users the flexibility to manage their visits according to their own needs and schedules while keeping everything organized and easy to track.

- View Patient History and Treatment Details

Authorized personnel, including dentists, clinic staff, and administrative Workers, have the ability to access and review a patient's full and complete history whenever required. This includes viewing records of past diagnosis, all treatments received in previous visits, procedures performed, medications used, and details of every appointment that has taken place. Having access to this comprehensive information is vital because it supports accurate and well-informed decision-making during current consultations, allowing medical professionals to understand the patient's background and make choices that are appropriate and effective. It also ensures continuity of care, meaning that even if different staff member or dentists are involved at different times, everyone has access to the same consistent information. This feature helps track how a patient's oral health has changed over time, identify patterns or issues that may need attention, and provide clear explanations to patients, about their current condition and treatment plans. By making this information easily accessible, the system improves the quality of

care, enhances communication between providers and patients, and helps ensure that all treatment is delivered with full knowledge of the patient's health status.

- Search and Retrieve Records

This feature enables users to quickly and efficiently locate specific information within the system, even when there is a large volume of data stored. Users can search for records using various filters and criteria, such as the patient's full name, unique identification number, contact details, specific dates of visits or treatments, or types of services received. The search tool is designed to deliver fast and accurate results, so the required information can be found and displayed within seconds. This functionality saves significant time and effort compared to manual searching or handling physical files, and it ensures that important information is always readily available whenever it is needed whether for immediate consultation, administrative tasks, or reporting purposes. It improves overall operational efficiency, reduces delays, and minimizes the risk of errors or missing information, making it easier for staff to manage large amounts of data and provide timely service to patients.

- Generate Reports (Appointments, Treatments)

This use case allows the system to automatically create organized, detailed, and easy-to-read summaries and documents based on the data stored within it. These reports can cover a wide range of information, including daily or monthly lists of appointments, statistics on the number and types of treatments performed, trends in patient visits, data on service usage, and other relevant operational or clinical information. These documents are extremely valuable for clinic management and administrators, as they provide clear insights into how the clinic is functioning, which services are most used, and how resources are being utilized. By analyzing these reports, management can make informed decisions about scheduling, staffing, service improvements, and overall business strategies. Additionally, these reports serve as important records for compliance, evaluation, and reference purposes, helping the clinic monitor its performance, identify areas where improvements can be made, and ensure that services are delivered effectively and efficiently to all patients.

- Manage User Access and Security (Admin)

Assigned specifically to the system administrator, this use case covers all activities related to controlling who can use the system and what information they can access or modify. It includes creating new user accounts for staff members, assigning specific roles and levels of permission to each account, updating login credentials and security settings, and defining exactly which functions and data each user is allowed to interact with. This control is essential to ensure that sensitive and private patient information remains secure and protected; it ensures that only authorized individuals can view, edit, or delete personal and clinical data, and that information is not disclosed or altered by people who do not have permission to do so. The administrator also monitors system activity to detect any unusual behavior or potential security risks, and they perform regular checks to ensure that all access and security in this way, the system maintains the privacy and integrity of all data, complies with legal requirements and health information standards, and creates a safe and trustworthy environment for both patients and staff. Furthermore, the administrator is responsible for system maintenance activities such as

updating software components, fixing technical issues, and ensuring that the platform runs smoothly without interruptions. They also implement disaster recovery plans and ensure that data can be restored quickly in case of system failure. Additionally, they manage audit trails to track all user activities within the system, which helps in identifying unauthorized actions and maintaining transparency. This continuous oversight strengthens system reliability and ensures long-term operational stability.

End of Proposal